UNIVERSITÀ DEGLI STUDI DI MILANO
FACOLTÀ DI SCIENZE E TECNOLOGIE

Reinforcement-Learning Framework for Robust and Universal Analog Quantum Control on Superconducting Qubits

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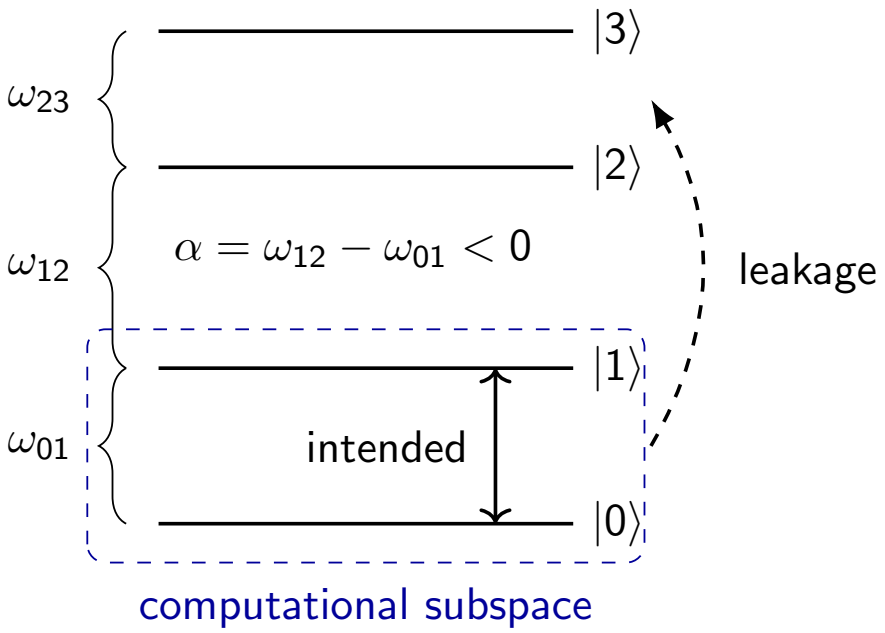
Context and qubit architecture

- For scalable and reliable quantum devices, high-fidelity, fast, and robust quantum gates are needed.
- Superconducting qubits are a leading platform, but makes control design challenging due to leakage and noise sensitivity.



Physical issue

Superconducting qubits are **weakly anharmonic**, so noise or fast control can easily excite unwanted higher levels. This phenomenon is called **leakage**.

Analog quantum control

Implementation of a quantum operation by injecting control pulses at the Hamiltonian level.

Why this problem matters

Main difficulty

In superconducting qubits, gate design is not only a matter of logical accuracy. We must balance several competing goals:

- **high gate fidelity**
- **short runtime**
- **low leakage** outside the computational subspace
- **robustness** against control noise

Why Deep Reinforcement Learning (RL)?

Thesis goals and main contribution

Scientific goal

To assess whether the framework proposed by Niu *et al.* is a credible strategy for **robust analog quantum control** on simulated gmon qubits.

What I specifically did

Significant effort was spent on implementation: no public code was available, the whole framework had to be rebuilt from scratch. This included:

- Quantum Environment simulator of two-qubit gmon architecture (QuTip)
- Leakage estimator
- Full Deep Reinforcement Learning pipeline (neural networks, training, evaluation), coupled with Trust Region Policy Optimization (pytorch)
- Optimization to make the framework computationally feasible
- Experiment design and independent validation

Physical model: gmon Hamiltonian and control channels

The effective Hamiltonian is

$$\hat{H}(t) = \underbrace{\frac{\eta}{2}\sum_{j=1}^{2}\hat{n}_j(\hat{n}_j - 1)}_{\text{anharmonicity}} + \underbrace{g(t)(\hat{a}_2^\dagger \hat{a}_1 + \hat{a}_1^\dagger \hat{a}_2)}_{\text{tunable coupling}} + \underbrace{\sum_{j=1}^{2}\delta_j(t)\hat{n}_j}_{\text{local detuning}} + \underbrace{\sum_{j=1}^{2} i f_j(t)\left(\hat{a}_j e^{i\phi_j(t)} - \hat{a}_j^\dagger e^{-i\phi_j(t)}\right)}_{\text{microwave drive}}$$

Control vector: Agent's output

The control vector is:

$$a_t = \big(g(t),\, \delta_1(t), \delta_2(t),\, f_1(t), f_2(t),\, \phi_1(t), \phi_2(t)\big)$$

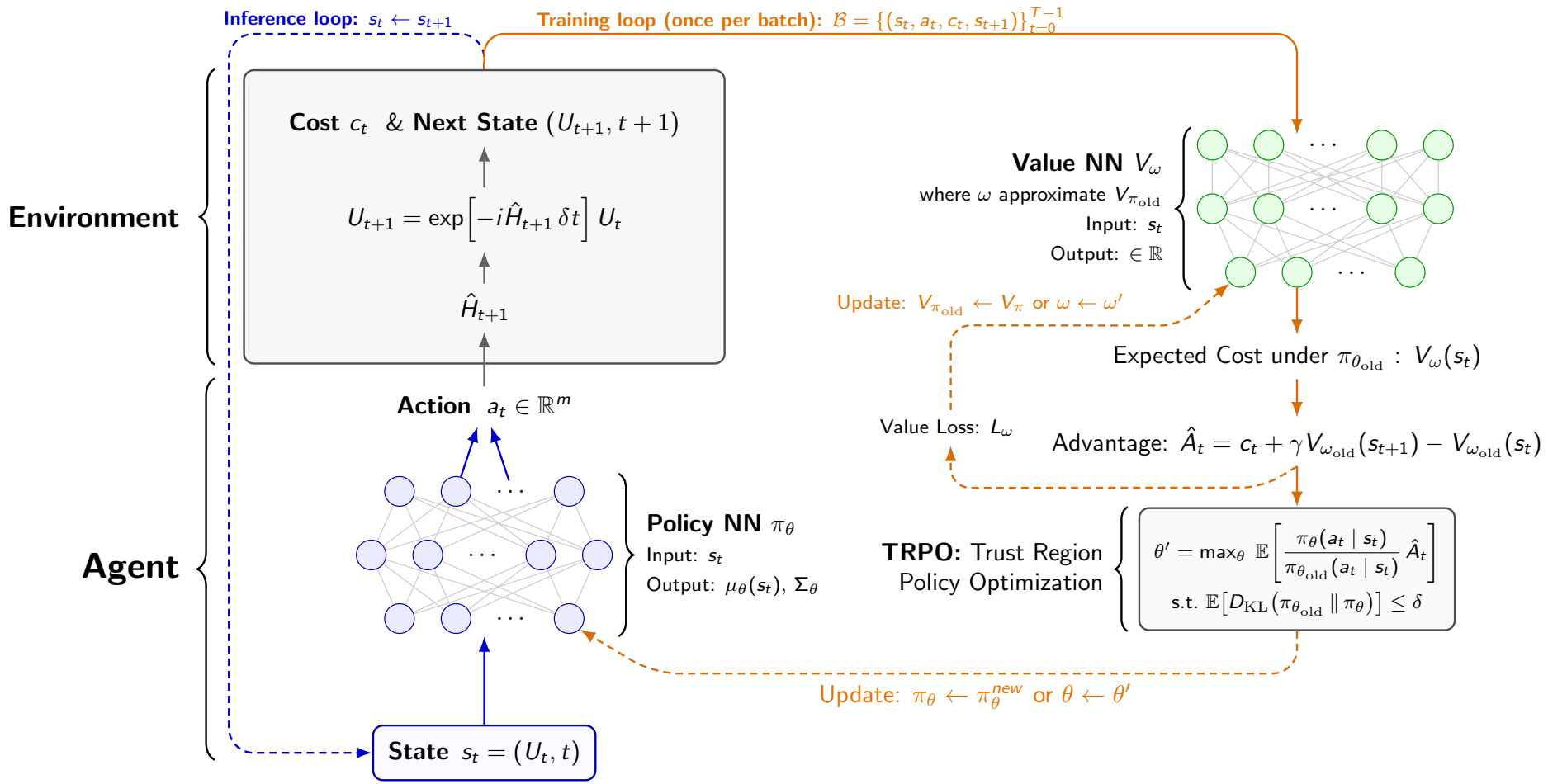
And this is going to be the output of the RL agent at each time step.

Solution to the control problem

A solution to the control problem is a sequence of control vectors

$$\{a_t\}_{t=0}^{T}$$

RL formulation



Why this is a multi-objective control problem

Cost Function and Multi-objective nature

The cost function is the core metric that the agent is trained to minimize. Therefore, the multi-objective nature of the control problem must be explicitly encoded in the cost function.

Universal Function Optimizaion (UFO) cost function

$$C = \underbrace{\chi[1 - F(U(T))]}_{\text{fidelity penalty}} + \underbrace{\beta\, L_{\mathrm{tot}}(T)}_{\text{leakage penalty}} + \underbrace{\mu\, P_{\mathrm{bdry}}}_{\text{boundary penalty}} + \underbrace{\kappa\, T}_{\text{time penalty}}$$

- These goals are not automatically compatible: reducing one penalty can worsen another.

Experiments performed

Controller variants implemented

- **Classical Optimization**: gradient descent-based pulse optimization (Adam) of the UFO cost. It's used as the baseline for comparison.
- **Reinforcement-Learning Agent** trained in multiple configurations.

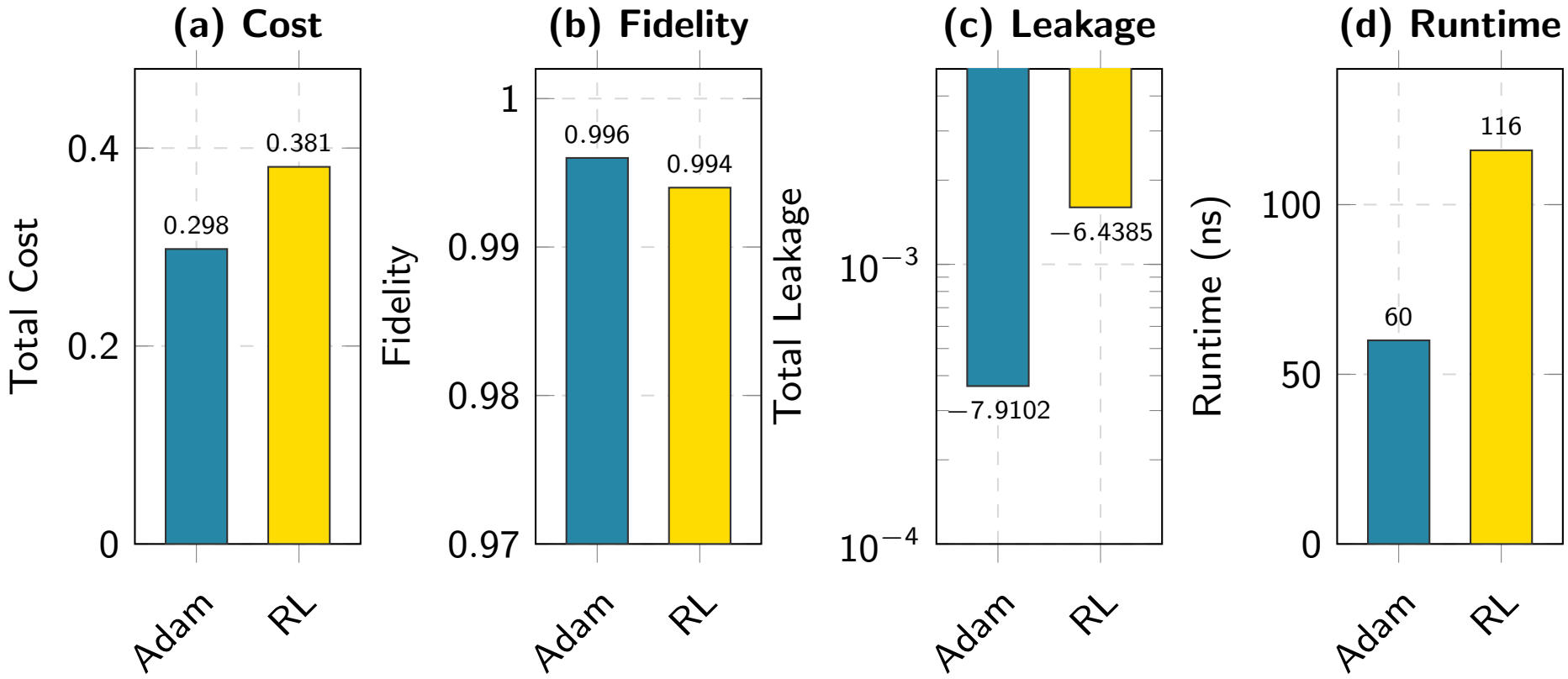
Main experiments performed

1. **Extensive parameter study**:
   - **neural network sizes and layouts**
   - **UFO cost function weights**
2. **Single-target benchmark**: Adam vs RL Agent
3. **Robustness under noise**: fidelity metrics vs noise strength
4. **Consistency of performance**: RL Agent vs Adam optimization
5. **Transfer Learning Evaluation**: across nearby targets

Single-target optimization: Nominal RL agent vs Adam

Interpretation

In the nominal single-target regime, the direct Adam baseline remains stronger.

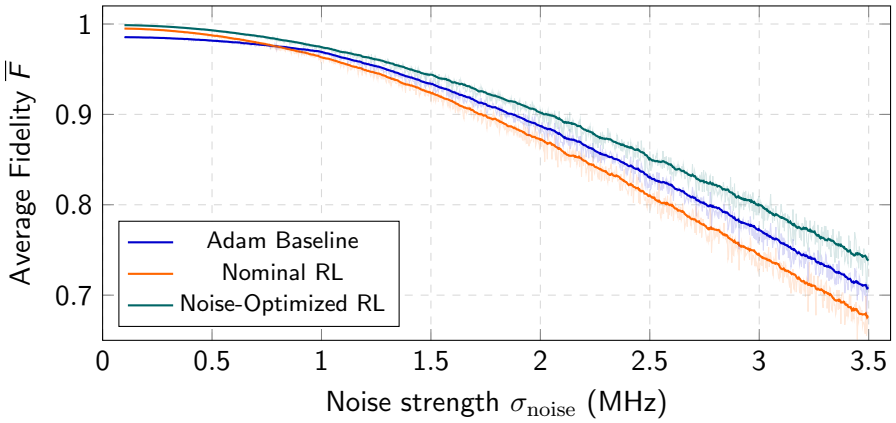


Noise-optimized Agent: Robustness

Under control noise, the **noise-optimized RL agent** (trained with 1MHz noise) is the strongest. The key advantage is not just RL, but **RL trained in stochastic environment**.

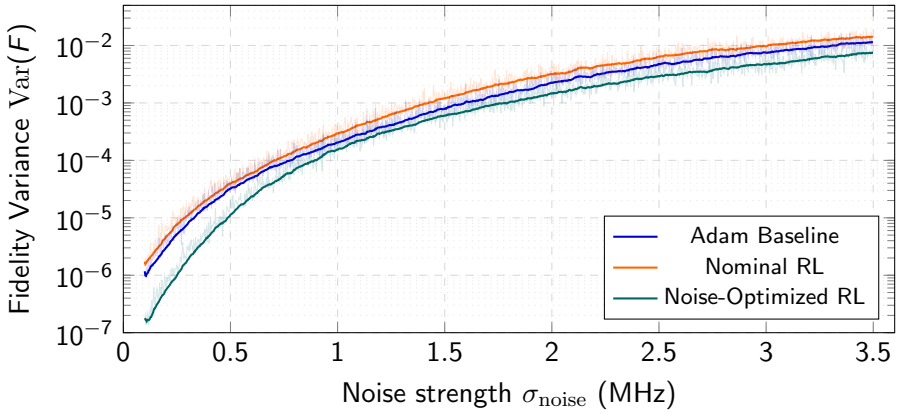
Average fidelity

- retains highest $\overline{F}(\sigma_{\text{noise}})$
- outperforms Adam and nominal RL



Fidelity variance

- demonstrates lowest $\mathrm{Var}(F)$
- superior structural stability under noise



Conclusions

What is not fully shown in this presentation

Although I only presented the main results, the thesis also includes several additional analyses:

- extended robustness comparisons under stochastic control noise
- runtime analysis across the gate family for both the baseline and the RL controller
- sensitivity studies on the UFO cost-function weights
- neural-network architecture sweeps
- transfer-learning behavior under curriculum training
- exploration of training and simulation parameter choices

What this thesis contributed

- full reconstruction of the framework from scratch
- extended existing literature with a comprehensive comparison between RL and a strong classical technique.

Scientific conclusions

- Under stochastic perturbations, noise-optimized RL becomes the best strategy.

Additional materials and results follow

Additional work beyond what's shown here

What I had to build and learn

To carry out this thesis, I had to reconstruct the full framework from scratch and acquire working knowledge of:

- QuTiP and multilevel quantum simulation, pytorch, autograd
- classical optimization and both classical and deep reinforcement learning
- policy-gradient methods and neural-network training
- superconducting-qubit physics and quantum control

UFO terms insights

Universal Function Optimizaion (UFO) cost function

$$C = \underbrace{\chi[1 - F(U(T))]}_{\text{fidelity penalty}} + \underbrace{\beta\, L_{\text{tot}}(T)}_{\text{leakage penalty}} + \underbrace{\mu\, P_{\text{bdry}}}_{\text{boundary penalty}} + \underbrace{\kappa\, T}_{\text{time penalty}}$$

Fidelity

Being $U(T)$ the unitary evolution generated by a control protocol at final time T, and U_{target} the desired logical gate.

$$F(U(T)) = \frac{1}{d_{\text{comp}}^2} \left| \text{Tr}\left(U^\dagger(T)\, U_{\text{target}} \right) \right|^2, \qquad \text{with} \quad d_{\text{comp}} = 4$$

Boundary Penalty

$$P_{\text{bdry}} = \sum_{t \in \{0, T\}} \left[g(t)^2 + f_1(t)^2 + f_2(t)^2 \right]$$

where $g(t), f_1(t), f_2(t)$ are the amplitude controllers at a given time t. This term forces (by penalizing) the control amplitude to be zero at the pulse boundaries, which can cause unwanted excitations and leakage after the termination of the episode.

UFO terms insights

Leakage term: physics-informed proxy

Considering the separated Hamiltonian in block form, $\hat{H}_{\mathrm{od}}(t)$ is the off-diagonal block that couples the computational subspace to the leakage subspace, while $\Delta(t)$ is the instantaneous energy gap between these subspaces.

$$L_{\mathrm{tot}}(T) \lesssim \frac{\|\hat{H}_{\mathrm{od}}(0)\|}{\Delta(0)} + \frac{\|\hat{H}_{\mathrm{od}}(T)\|}{\Delta(T)} + \int_0^T \frac{1}{\Delta(t)^2} \left\| \frac{d^2 \hat{H}_{\mathrm{od}}(t)}{dt^2} \right\| dt$$

The first two terms penalize residual couplings at the pulse boundaries; the integral term penalizes rapid temporal variations that can induce non-adiabatic transitions, effectively working as a smoothing filter.

Time penalty

Since the experiment values do not coincide with the natural time units of the system, a normalization factor is introduced.

$$T = \frac{t}{T_{\mathrm{norm}}}, \quad \text{with } T_{\mathrm{norm}} = 60\,\mathrm{ns}$$

Choosen target gate family

Target gate family used throughout the numerical study.

$$N(\alpha, \alpha, \gamma) = \exp\left[i\left(\alpha\,\sigma_1^x\sigma_2^x + \alpha\,\sigma_1^y\sigma_2^y + \gamma\,\sigma_1^z\sigma_2^z\right)\right].$$

Why...

Evaluation metrics & target gate family

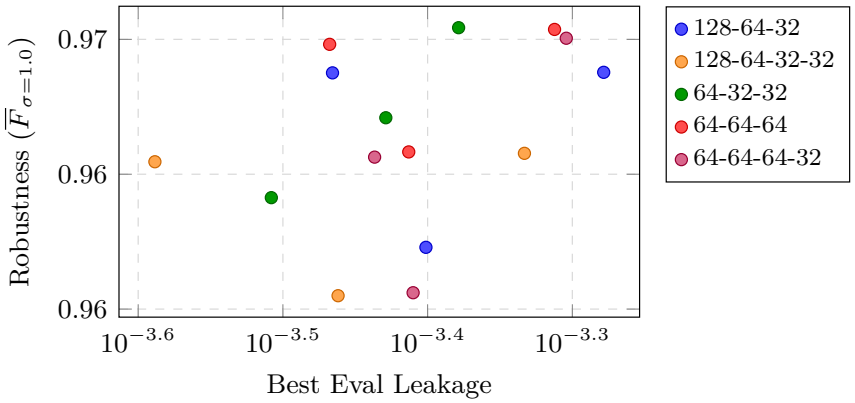
Metric	Definition	Physical meaning
Gate fidelity	$F_{\mathrm{gate}}=\frac{1}{16}\left\lvert \mathrm{Tr}\left(U^{\dagger}_{\mathrm{proj}}(T)U_{\mathrm{target}}\right)\right\rvert^2$	Logical accuracy of the implemented two-qubit gate.
Leakage proxy	$L_{\mathrm{tot}}(T)\lesssim\frac{\lVert\hat{H}_{\mathrm{od}}(0)\rVert}{\Delta(0)}+\frac{\lVert\hat{H}_{\mathrm{od}}(T)\rVert}{\Delta(T)}+\int_0^T\frac{1}{\Delta(t)^2}\left\lVert\frac{d^2\hat{H}_{\mathrm{od}}(t)}{dt^2}\right\rVert dt$	Estimated coherent population loss outside the computational subspace.
Average fidelity under noise	$\overline{F}(\sigma)=\mathbb{E}[F]$	Mean gate performance over many noisy Monte Carlo realizations.
Fidelity variance	$\mathrm{Var}(F)=\mathbb{E}\left[(F-\overline{F})^2\right]$	Shot-to-shot stability under stochastic perturbations.

Extended parameter study

In RL, parameter choice strongly affects stability, convergence, robustness, and final control quality. I performed an extended study on:

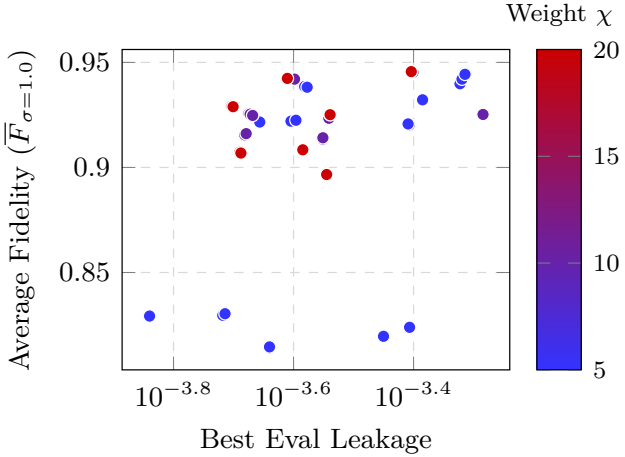
Policy/value architectures

- sweep over 5 network topologies
- final choice: $(64, 32, 32)$



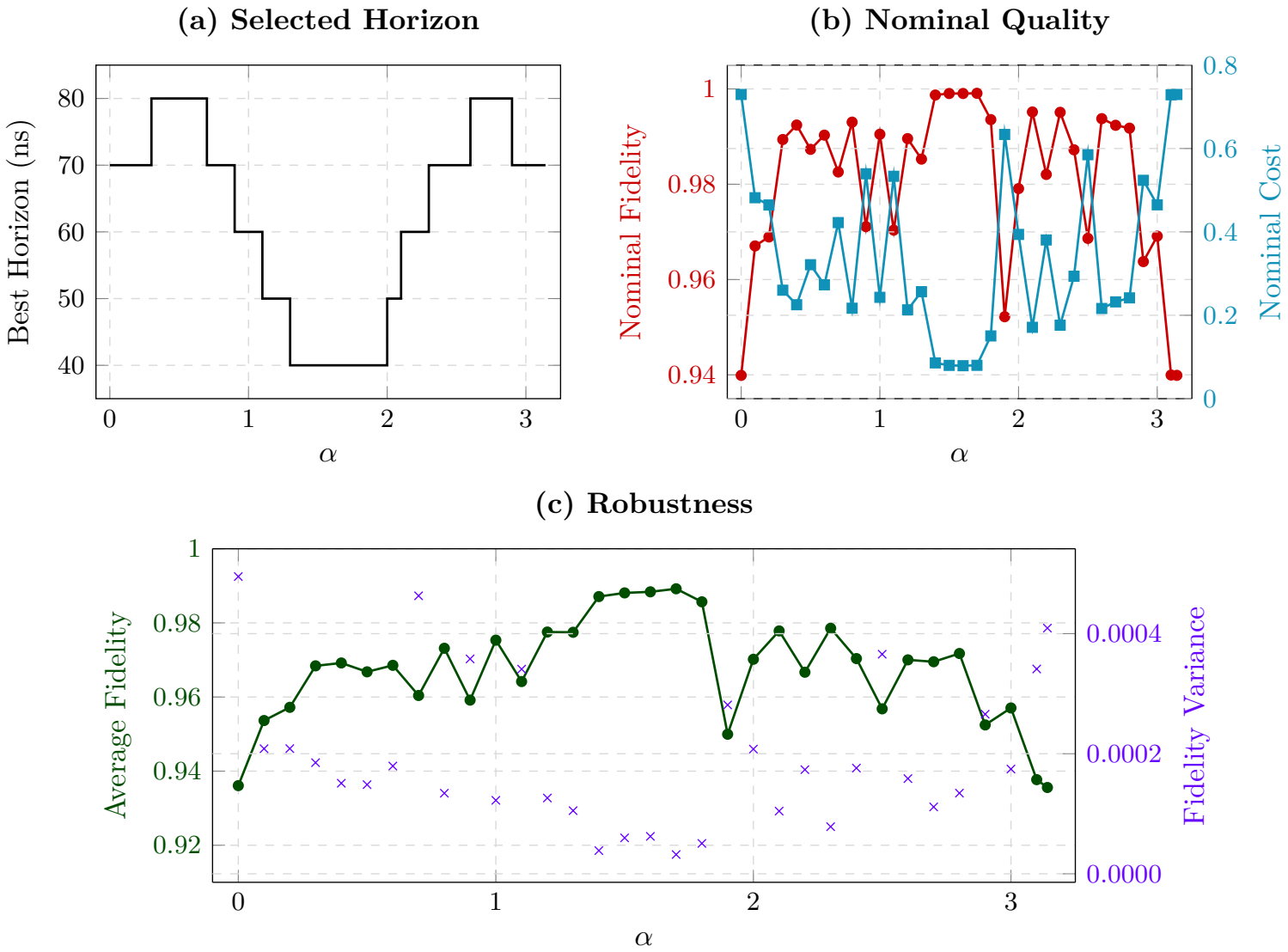
UFO weights

- sensitivity study over 36 configurations
- final choice: $(10, 10, 0.2, 0.1)$



Adam vs TRPO and Family-wide evaluation

Family-wide runtime sweep was also evaluated to mostly assess performance consistency across the target gate family of the Adam baseline.



Runtime across the gate family $N(\alpha, \alpha, \pi/2)$

Target parametrization: $U_{\text{target}}(\alpha) = N(\alpha, \alpha, \pi/2)$. The runtime landscape is **highly structured**, not uniform.

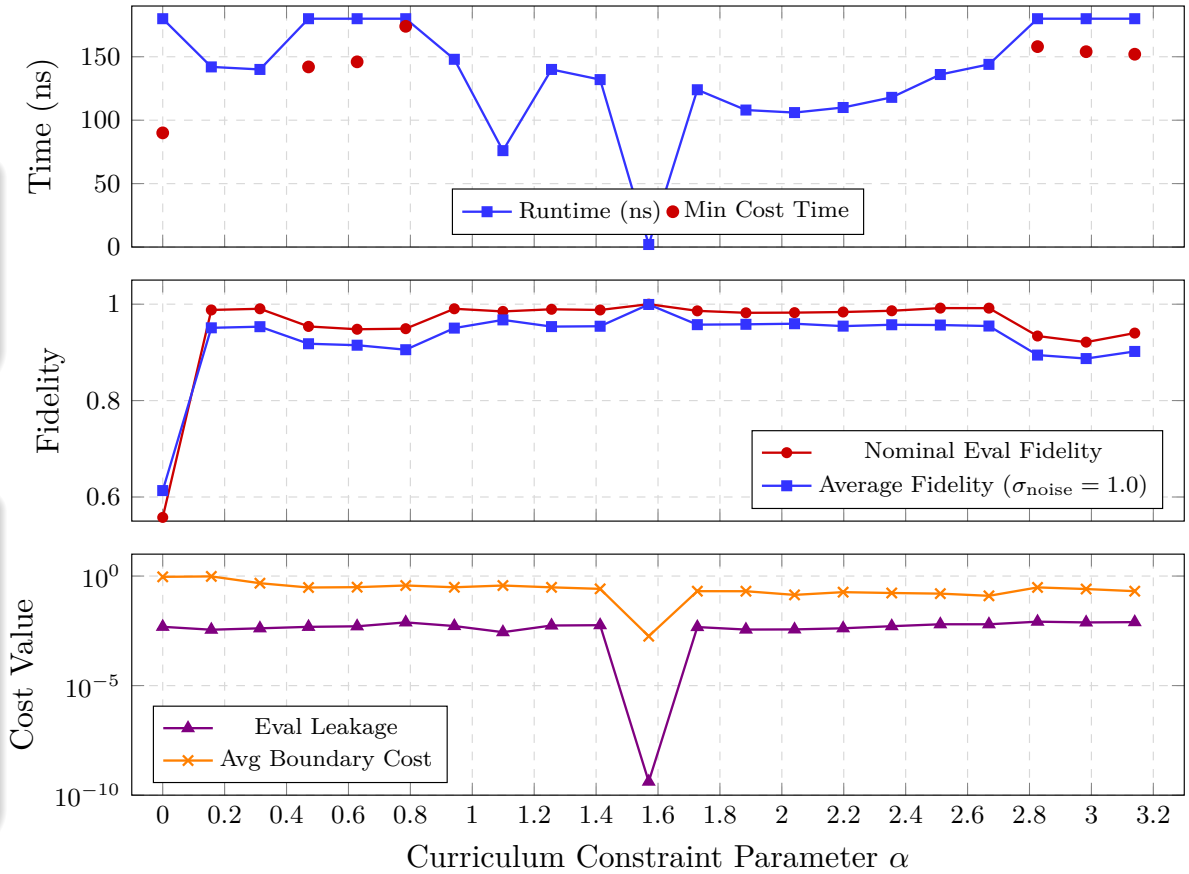
- A broad central region admits significantly faster analog solutions.
- The analog controller stays below the gate-synthesis reference over a large part of the family.

Reference benchmark

$T_{\text{syn}} \approx 215\,\text{ns}$ from optimal discrete gate synthesis.

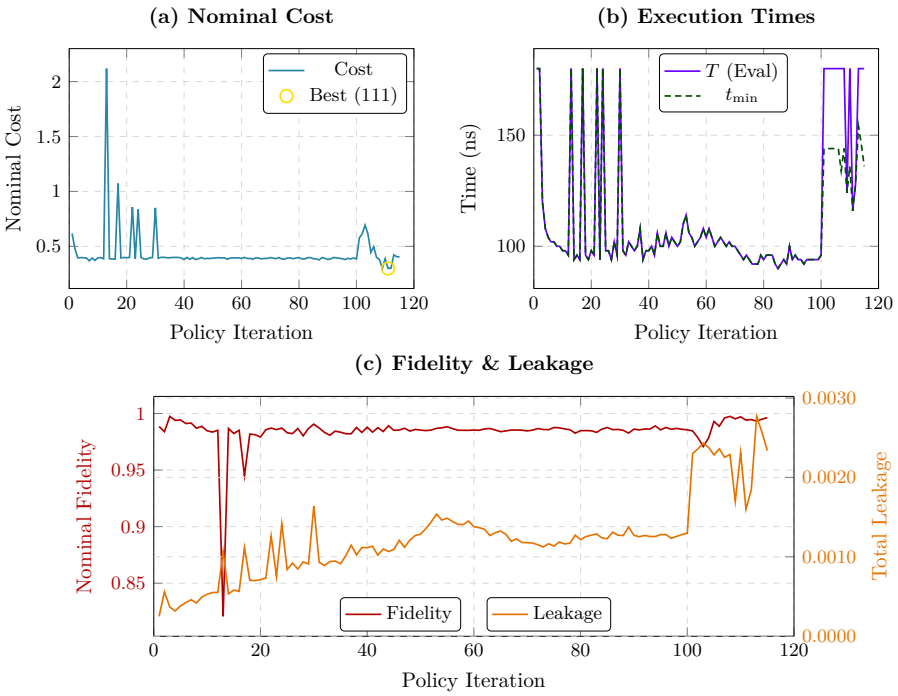
Physical interpretation

The strongest speedups appear where the target gate is better aligned with the native hardware interaction.

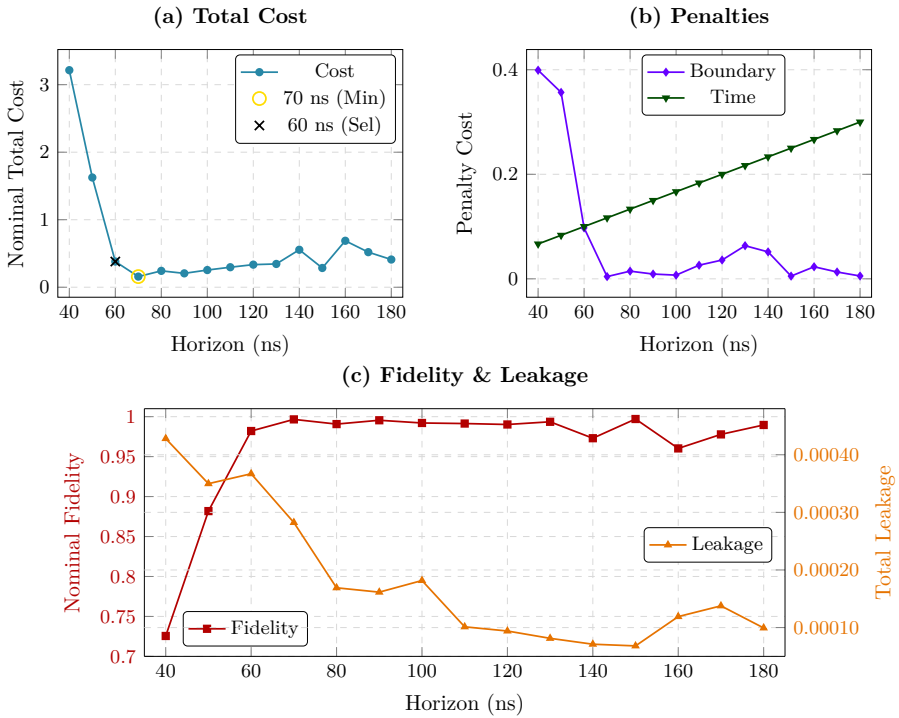


Single-target: Training Dynamics and Horizon Sweep

RL Agent training dynamics



Single-target horizon sweep



RL state

Current propagator and state definition

The current propagated unitary is

$$U_t = \mathcal{T} \exp\left(-i \int_0^t H(\tau)\, d\tau\right),$$

so it represents the **time-evolution operator accumulated up to time** t.

In the implementation, the RL state is built from the current propagator as

$$s_t = \Big(\mathrm{Re}(\mathrm{vec}(U_t)), \mathrm{Im}(\mathrm{vec}(U_t)), t/T_{norm}\Big).$$

With a three-level truncation for each mode:

$$d_{\mathrm{phys}} = 3^2 = 9 \quad \Longrightarrow \quad U_t \in \mathbb{C}^{9\times 9},$$

so the observation dimension is

$$d_{\mathrm{obs}} = 2 \cdot 9^2 + 1 = 163.$$

Policy

Policy network

Input: current state s_t

Output: parameters of a Gaussian policy

$$\pi_\theta(a_t \mid s_t) = \mathcal{N}\big(\mu_\theta(s_t),\, \Sigma_\theta\big),$$

with

$$\mu_\theta(s_t) \in \mathbb{R}^7, \qquad \Sigma_\theta = \mathrm{diag}(\sigma_1^2, \ldots, \sigma_7^2).$$

After sampling and squashing:

$$a_t = \tanh(\tilde{a}_t) \in [-1, 1]^7,$$

then rescaled to the physical controls

$$u_t = \big(g_t, \delta_{1,t}, \delta_{2,t}, f_{1,t}, f_{2,t}, \phi_{1,t}, \phi_{2,t}\big).$$

Value network

Value network

Input: current state s_t

Output: scalar state value

$$V_\omega(s_t) \in \mathbb{R}.$$

This estimates the expected discounted future reward:

$$V_\omega(s_t) \approx \mathbb{E}\left[\sum_{k=t}^{T} \gamma^{k-t} r_k \,\middle|\, s_t\right].$$

Interpretation:

- the **policy** proposes the next control action,
- the **value network** estimates how promising the current state is.